

Local goodness-of-fit measures for neural decoding

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Abstract

State-space models are widely used in many fields to infer latent states from observations, but there are still some issues and challenges related to them. First, because latent variables are inherently unobservable, their decoding results cannot be directly validated against observed data. Second, it is often challenging to disentangle prediction errors stemming from model misspecification. Third, such errors can accumulate and propagate over time, amplifying uncertainty in long-term predictions. Moreover, many state-space models are purposefully simplified, accepting some degree of inaccuracy in exchange for greater computational efficiency. To address this challenge, we developed three classes of metrics designed to identify localized periods where model misspecification leads to inaccurate estimates. Each metric assesses the consistency between the information contained in individual observations and the information extracted by the state-space model from a local window of data. These metrics fall into three categories: 1) a set of divergence measures, specifically

F-divergences, which compare the distributions of the latent state process directly; 2) measures of overlap between estimated credible regions obtained from the state distributions; and 3) predictive checks for individual observations, based on samples from the estimated state process. We applied these metrics to the problem of estimating nonlocal spatial representations in both simulated and real hippocampal spiking data, demonstrating the strengths and limitations of our approach.

Keywords: Neural decoding, State-space models, Goodness-of-fit

1 Introduction

Many brain states, such as those related to memory, emotion, and attention, are not directly observable. They are internal states that are not solely determined by the external world and can change over time. This makes them challenging to understand. However, these brain states are critical for understanding how the brain supports complex behavior, because they go beyond simple stimulus and response relationships. A major challenge for systems neuroscience is to relate neural activity (spikes, spike waveforms, calcium traces, etc.) to these latent dynamic brain states.

State space models are a powerful way to understand how neural activity relates to latent dynamic brain states. They have been applied to diverse neural systems such as gustatory activity [Jones et al., 2007], hippocampal replay dynamics [Denovellis et al., 2021], and hypothalamic activity dynamics [Vinograd et al., 2024]. The core assumption of these state space models is that complex, high dimensional neural activity can be related to low dimensional, latent states. State space models aggregate information over time about the latent states, weighing both the information from the data at the current time (normalized likelihood distribution) with the accumulated information used to predict the latent state (prediction distribution). This powerful combination enables robust estimation of the latent state, effectively balancing current information with the accumulated history of the system and prior knowledge.

Despite the wide use of state space models in neuroscience, methods to validate state space models remain underdeveloped and underutilized. One reason is that latent states are inherently

difficult to evaluate [Newman et al., 2014]. Without ground truth, model validity can only be inferred indirectly—from relative likelihoods under multiple models, from the quality of reconstructed signals, or from performance on downstream tasks. Another challenge is that neural data are dynamic, so model fit may vary substantially across time. Compounding this problem is that common methods of validating models, such as cross-validation and “shuffle” permutation tests, are holistic evaluations of the model – they tell neuroscientists whether or not the model generalizes well or is different from a simplified model with less utility – but they fail to indicate when the model fits poorly or which specific components are responsible. Consequently, these methods can reject models that are intentionally simplified yet scientifically meaningful and offer little guidance for targeted improvement of the model. Many state-space models in neuroscience are intentionally simplified—not because the brain is simple, but because tractable computation and clear interpretation often matter most. For example, some state-space models smooth emotional movement into a low-dimensional trajectory [Deng et al., 2015]; others compress field-linked structure into a stationary, locally anchored map [Hsin et al., 2022]. Some frameworks simplify both at once, jointly compressing affective trajectories and local field potentials into compact latent states for computation and inference [Tao et al., 2018, Zeng and Eden, 2025]. Under standard goodness-of-fit criteria, such simplifications would likely fail.

State space models already contain rich diagnostic information in the form of distributions used in their estimation. One potential solution for more targeted local diagnostic information of state space models would be to utilize the information in the normalized likelihood, which reflects information from only the current neural observations, and prediction distribution, which aggregates information from all past observations based on the state space model structure. When these two distributions agree, the model’s prior expectations and data-driven evidence are consistent. When they diverge, the mismatch can reveal where and when and potentially how the model fails to capture the structure of the data. Examining these discrepancies could therefore provide a direct, interpretable measure of model–data agreement—analogueous to a local, time-resolved goodness-of-fit test.

To address these gaps, we develop and validate a suite of three complementary goodness-of-fit methods – KL divergence, highest posterior density (HPD) overlap region, predictive checks – designed to provide actionable feedback to neuroscientists. Specifically, each of these methods utilizes the mismatch between prediction and likelihood to: (1) identify time periods of poor model–data agreement; (2) distinguish errors arising from the state versus observation components; and (3) provide intuitive visualizations accessible to a broad audience. We demonstrate the utility of these techniques on simulated and real hippocampal data, offering a path for neuroscientists to move from binary model rejection toward an iterative, diagnostic approach to building interpretable theories of neural dynamics.

2 Methods

2.1 State-Space Models

The state-space paradigm is particularly well suited for neural data analysis since many neuroscience experiments focus on internal brain processes that are not directly observable. For example, hippocampal replay of spatial trajectories has been hypothesized to be related to mental exploration of space, but the specific spatial trajectories being replayed are not directly observable by experimentalists [Tao et al., 2018]. In such cases, state-space models are used to estimate simultaneously the unobserved internal representations and their relation to neural activity patterns.

For a discrete set of time points, $t_0, \dots, t_K$, we define an unobserved state process x_k , which influences a neural observation process y_k at each time t_k . We define a state space model by combining a state transition distribution $p(x_k | x_{k-1})$, which defines how the state evolves in time, with an observation likelihood $p(y_k | x_k)$, which defines the neural representation of each state. To simplify the problem, we assume that the state transition depends only on the most recent past value of the state and that the observation likelihood depends only on the current state value. Figure 1(a) shows a schematic of this state-space model structure.

This state-space model structure allows for iterative estimation of the state process at each time

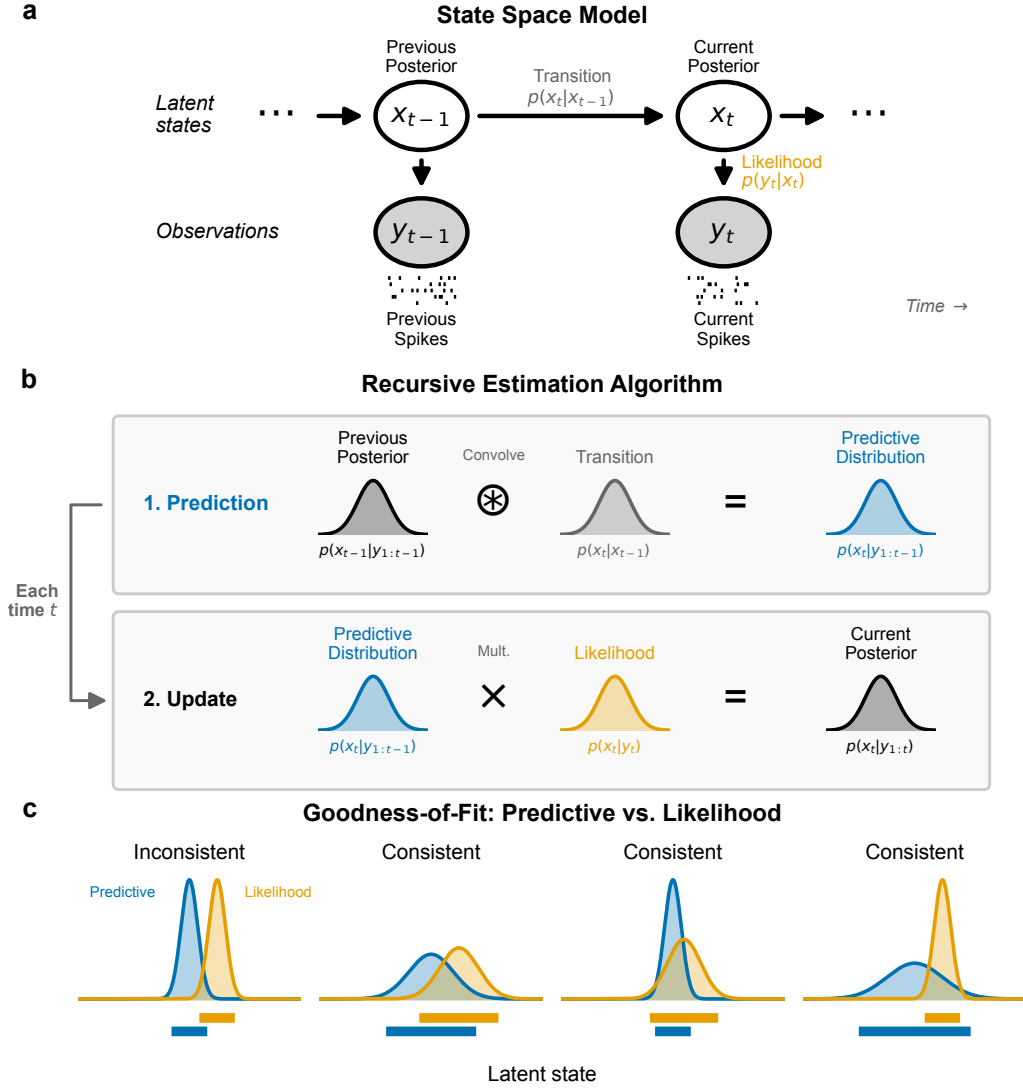


Figure 1: Schematics for state-space model and goodness-of-fit measures. (a) Graphical model depiction of state-space model includes state transition model, $p(x_k | x_{k-1})$, and observation likelihood $p(y_k | x_k)$. (b) These model components are sufficient to compute an iterative expression for the posterior filter distribution at time t_k , $p(x_k | y_{1:k})$ from the distribution at the previous time step. This involves first convolving the previous posterior with the state transition distribution to compute the one-step predictive distribution, $p(x_k | y_{1:k-1})$, and then updating the predictive distribution with information from the most recent spiking by multiplying by the observation likelihood. (c) For each new spike observation, we compute the local goodness-of-fit by assessing the consistency of information between the predictive distribution and likelihood. These are considered inconsistent when they assign high probability to mostly disjoint subsets of the state-space. When the regions with high probability overlap, the distributions are considered consistent, even if the distributions differ in their central tendency or certainty.

step based on all the neural data up to that time, as shown in Figure 1(b). If the distribution of the state at time t_{k-1} has been computed, we can convolve it with the state transition distribution to obtain a one-step prediction distribution, $p(x_k | y_{1:k-1}) = \int_{x_{k-1}} p(x_k | x_{k-1}) \cdot p(x_{k-1} | y_{1:k-1}) dx_{k-1}$. This predictive distribution combines the information from all previous observations to compute the distribution of the state at the next time step. We update this predictive distribution with the information from the most recent observation by multiplying by the observation likelihood to compute the posterior distribution of the state given all of the data up to the current time step, $p(x_k | y_{1:k}) \propto p(y_k | x_k) \cdot p(x_k | y_{1:k-1})$.

Common approaches for assessing the goodness-of-fit of state-space models include examining the cross-validated error measures in the state process when the state is observable [Stone, 1974], cross-validated error measures in the observation process such as the mean-squared prediction error [Hastie et al., 2009], and the distribution of prediction residuals [Shen and Zhuang, 2024]. For many neural modeling problems, state-space models are intentionally simplified and these standard goodness-of-fit measures trivially show lack of fit.

Here we explore a set of local goodness-of-fit measures, each of which compares the prediction distribution to the normalized likelihood of the most recent observation as a function of the state or to that observation itself. Thus, these measures consider the consistency between information accumulated through the state-space model over past spiking with the information from the most recent spiking activity. We do not expect these distributions to align perfectly since they likely contain different amounts of information about the state process. Instead, we assess whether the information these distributions provide about the state and observation processes is consistent in the sense illustrated in Figure 1(c). Specifically, we say that the two distributions are inconsistent when their probability mass is concentrated in mostly non-overlapping regions of the state space. This suggests disagreement between the information from past spiking and the information from the most recent spiking. Distributions that largely align, or that are shifted from each other but are supported over largely overlapping regions of state space are considered consistent. Additionally, if one distribution covers a broad region of the state-space and the other covers a much narrower subset

of that region, the distributions are considered to be consistent. We expect this situation to arise often since the prediction distribution contains information accumulated from many past spikes.

2.2 Local Goodness-Of-Fit Measures

In this section, we introduce three classes of metrics designed to characterize the local goodness-of-fit of the state-space model.

2.2.1 KL Divergence

One of the most well-known measures of discrepancy between probability distributions is the Kullback–Leibler (KL) divergence [Kullback and Leibler, 1951], defined in this case as the expected excess self-information (log of the density function) incurred when the likelihood is used to approximate the predictive distribution, with the expectation taken with respect to the predictive distribution,

$$D_{\text{KL}}(P \parallel Q) = \int P(x_k) \log\left(\frac{P(x_k)}{Q(x_k)}\right) dx_k, \quad (1)$$

where $P(x_k) = p(x_k \mid y_{1:k-1})$ is the predictive distribution and $Q(x_k) = p(y_k \mid x_k) / \int p(y_k \mid x_k) dx_k$ is the normalized likelihood in x_k given the observed spiking y_k .

Figure 2(a) illustrates the computation of the KL divergence. The top panel shows the predictive distribution and normalized observation likelihood as functions of x_k . The middle panel shows the difference in self-information between these distributions, $\log(P(x_k)/Q(x_k))$. The bottom panel is this difference in self-information weighted by $P(x_k)$, which when integrated gives the value of the KL divergence.

The KL divergence is an asymmetric measure, meaning that $D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$. We could also compute the KL divergence from the likelihood to the predictive distribution $D_{\text{KL}}(Q \parallel P)$ or the average of these two divergences, $\frac{1}{2}(D_{\text{KL}}(P \parallel Q) + D_{\text{KL}}(Q \parallel P))$. More broadly, the KL divergence is part of a larger family of divergence metrics between distributions, known as F-divergences. Other members include the total variation distance [Bhattacharyya et al., 2023] and the

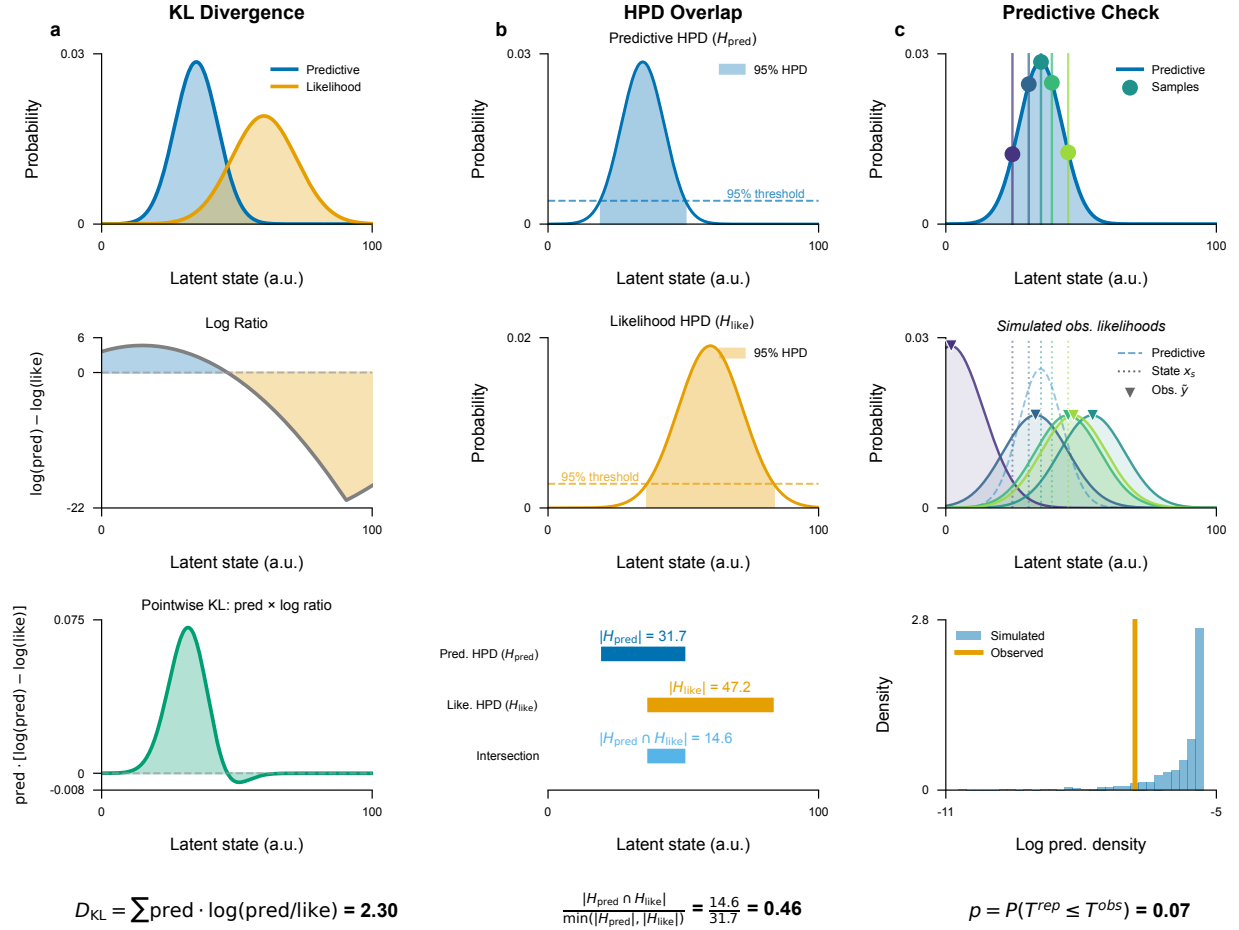


Figure 2: Visualizations of the computation of each local goodness-of-fit metric. (a) The KL divergence compares the predictive distribution (blue) and the observation likelihood (orange) (top panel) by computing the difference in their log density (middle panel) and computing its expectation with respect to the predictive distribution (bottom panel). (b) The HPD overlap is defined by computing the HPD for the predictive distribution (top) and for the observation likelihood (middle) and then computing the size of their intersection relative to the size of the smaller of these two HPD regions (bottom). (c) Predictive checks are computed by sampling values of the state from the predictive distribution multiple times (top), then sampling from the conditional distribution of neurons that could have generated the observed spike given each sampled state value (middle), and then computing the probability that the log of the predictive distribution of the observed spike is greater than the log of empirical predictive distribution of the sampled spikes (bottom).

Pearson χ^2 divergence [Crack, 2018]. It is simple to extend the KL divergence metric between the predictive distribution and the observation likelihood to any of these other divergence metrics to identify differences in additional features of the distributions.

2.2.2 Overlap Between HPD Regions

Another way to assess the consistency between the predictive distribution and the observation likelihood is to examine the extent to which their highest posterior density (HPD) regions overlap. A substantial intersection suggests that the two distributions convey consistent information, reinforcing the reliability of the inferred latent state. In contrast, minimal or no overlap signals a significant discrepancy, hinting at potential inconsistencies in the local fit of the model. This approach provides an interpretable and intuitive metric of the degree of consistency between these distributions.

We define the overlap between the HPD for the predictive distribution and the likelihood as

$$\text{HPDOverlap}(P, Q) = \frac{|\text{HPD}_P \cap \text{HPD}_Q|}{\min\{|\text{HPD}_P|, |\text{HPD}_Q|\}}, \quad (2)$$

where HPD_P and HPD_Q are the highest posterior density regions of the predictive density and the normalized observation likelihood respectively, and $|\cdot|$ denotes the volume of the region. The numerator is the size of the overlap between these regions and the denominator is the size of the smaller of the two HPD regions. This normalization ensures that the resulting value is in the range $[0, 1]$, with 0 indicating no overlap, and 1 indicating that one of the HPDs is entirely contained within the other. This means that if, for example, the likelihood produces an HPD that is wider and therefore less informative than the predictive density but that completely contains the HPD of the predictive density then the HPD ratio will be 1, suggesting that the two distributions are statistically consistent. Conversely, if the ratio is close to 0, it indicates little or no overlap, suggesting significant differences between the range of values for the state that would be inferred from these two distributions.

Figure 2(b) illustrates the computation of the HPD overlap. The HPD for the predictive

distribution (top) and the normalized likelihood (middle) are computed, and the size of their intersection is normalized by the size of the smaller of the two HPDs (bottom).

2.2.3 Predictive Checks

The previous metrics compared two distributions of the state: one based on the accumulated past information and the other based on the isolated information from the likelihood of the most recent spiking observation. Another approach is predictive checking, which compares the observation itself to a predictive distribution of that observation estimated from the model. [García and Castellanos \[2007\]](#) proposed Bayesian checking of the second levels of hierarchical models with partial posterior predictive checking. [Moran et al. \[2019\]](#) used population predictive checking to evaluate model’s goodness-of-fit. [Guttman \[1967\]](#) used future observations to evaluate model’s goodness-of-fit. Predictive checks have the advantage that they allow for the computation of p -values for individual observations, which are easy to interpret [[Meng, 1994](#), [Gelman et al., 1996](#)].

Predictive checks for spiking data can be challenging to implement, since in sufficiently small intervals, any spikes are unlikely compared to no spiking at all. One approach to deal with this might be to consider patterns of spikes over longer fixed intervals over which multiple spikes would be expected. Here, we take a simpler approach of focusing only on spike times and measuring the likelihood that an observed spike comes from a particular neuron (for spike-sorted models) or has a particular set of waveform features (for clusterless models).

In particular, let $y_{k,j}$ be the identity of the neuron (spike-sorted model) or a spike waveform feature (clusterless model) for the j th spike occurring in the k th time bin. We define the predictive distribution of the observation based on the state model as

$$f_{\text{pred}}(y_{k,j}) = \int p(y_{k,j} \mid x_k) p(x_k \mid y_{1:k-1}) dx_k, \quad (3)$$

where $p(y_{k,j} \mid x_k)$ is the observation process and $p(x_k \mid y_{1:k-1})$ is the one-step prediction distribution of the current state given all previous spikes. We can sample new observations $\tilde{y}_{k,j}$ from this

predictive distribution, and compute the probability that the observed spike is at least as likely as the sampled spike:

$$p_{k,j} = \Pr_{\tilde{y}_{k,j} \sim f_{\text{pred}}} (\log f_{\text{pred}}(y_{k,j}) \geq \log f_{\text{pred}}(\tilde{y}_{k,j})) . \quad (4)$$

The log terms in Equation (4) are unnecessary, but suggest the method to estimate $p_{k,j}$. We sample x_k multiple times from the one-step prediction distribution, $p(x_k \mid y_{1:k-1})$, and use these to generate samples of the spike observations, $\tilde{y}_{k,j}$, from the predictive distribution. We then compute the fraction of these samples whose log likelihood is smaller than or equal to that of the observed spike.

Figure 2(c) illustrates the computation of a predictive p -value for spike-sorted data. Many values of the state are sampled from the predictive distribution (top), each of which is used to sample a neuron that could have generated the observed spike (middle). The log likelihood of the observed spike is then compared to an empirical cumulative distribution of the log likelihoods of the sampled spikes (bottom) leading to a goodness-of-fit metric that can be interpreted as a p -value for that individual spike. High p -values suggest that the observed spike is consistent with the predictive distribution of spikes and a p -value near zero suggests that the observed spike is unexpected based on the predictive distribution computed from the past spiking and the model.

3 Simulation Study

Validating local goodness-of-fit metrics on real neural data is fundamentally limited: when the truth is unobserved, a metric that flags a moment of model failure cannot be distinguished from one that flags a moment of unusual neural activity. We therefore evaluate the three diagnostics—KL divergence, HPD overlap, and the rank-based predictive p -value—on a controlled benchmark in which the ground-truth fault at each moment is known by construction. The benchmark simulates a hippocampal place-cell population on a one-dimensional track and runs a Bayesian state-space decoder over the full session, deliberately perturbing either the generative process or the decoder’s assumptions during four distinct windows. These four perturbations were chosen to highlight the distinction between model misfit that leads to decoding errors versus model misfit to which the

218 decoding estimates remain robust. This relates to our original goal of identifying local periods
 219 where the information from the current spiking is inconsistent with the aggregated information from
 220 our model based on past spiking.

221 **3.1 Generative Model**

222 The simulation runs a Bayesian filter over a 32-second session whose generative process and
 223 decoder coincide everywhere except inside four misfit windows. Each window perturbs one specific
 224 component of the matched model.

225 **Track and observation model.** The state space is a one-dimensional linear track of length $L = 100$
 226 arbitrary units (a.u.) discretized at unit spacing, $x \in \{0, 1, \dots, 100\}$. The simulated population
 227 consists of eleven place cells with Gaussian place fields centered at $\mu_c \in \{0, 10, \dots, 100\}$ and
 228 shared standard deviation $\sigma_{\text{pf}} = 10$ a.u. The expected spike rate of cell c at position x is

$$\lambda_c(x) = \alpha \phi(x \mid \mu_c, \sigma_{\text{pf}}^2), \quad (5)$$

229 where $\phi(\cdot \mid \mu, \sigma^2)$ is the Gaussian density and $\alpha = 5$ is a peak-rate scale factor. With a 1 ms
 230 simulation step, the peak rate is $\alpha/(\sigma_{\text{pf}}\sqrt{2\pi}) \approx 200$ Hz. Spike counts are drawn independently
 231 across cells at each step, $y_{c,t} \sim \text{Poisson}(\lambda_c(x_t))$.

232 **Position dynamics.** In baseline windows the animal’s position evolves as a Gaussian random walk
 233 with step standard deviation $\sigma_{\text{pred}} = 0.5$ a.u. The trajectory is reflected at 0 and 100 a.u. throughout
 234 the session, starts at $x_0 = 0$, and is continuous across phases: the final position of each phase
 235 becomes the initial position of the next.

236 **Phase schedule.** The 32-second simulation ($T = 32,000$ steps) is partitioned into eight contiguous
 237 phases: an opening baseline, four misfit windows, and three clean-recovery interludes that restore
 238 the matched-model configuration. Each misfit window perturbs either the generative process or the

239 decoder:

240 **Remap (steps 6,000–10,000; decoder side):** the decoder evaluates per-spike likelihoods against a
241 remapped place-field map in which three cell pairs ($0 \leftrightarrow 9$, $1 \leftrightarrow 8$, $2 \leftrightarrow 7$) have swapped
242 centers, so the decoder’s intended likelihood is at the wrong location for six of the eleven cells.
243 Spike generation is unchanged.

244 **History-dependent firing (steps 14,000–18,000; generative side):** spikes are generated with a
245 1 ms hard refractory period followed by a 2–10 ms post-spike burst window in which the
246 rate is multiplied by 3, consistent with the rough phenomenology of CA1 pyramidal-cell
247 refractory and burst windows; the decoder continues to assume independent Poisson firing.
248 The per-spike spatial likelihood for any individual spike is unchanged, so the misfit lives
249 entirely in the spike-train temporal joint distribution.

250 **Drift (steps 22,000–26,000; generative side):** the trajectory is generated with persistent velocity,
251 $v_t = 0.8 v_{t-1} + \eta_t$ and $x_t = x_{t-1} + v_t$ with $\eta_t \sim \mathcal{N}(0, \sigma_{\text{pred}}^2)$; the decoder continues to assume
252 a memoryless random walk with step standard deviation σ_{pred} .

253 **Wide-dynamics noise (steps 30,000–32,000; decoder side):** the decoder is given an inflated tran-
254 sition matrix with step standard deviation $\sigma_{\text{inflated}} = 20$ a.u. (forty times the baseline), so its
255 one-step predictive distribution spreads broadly across the track; spike generation is unchanged.
256 This phase is engineered to inflate the KL divergence while leaving HPD overlap and the
257 rank-based predictive p -value near baseline.

258 The three intermediate clean-recovery windows (steps 10,000–14,000; 18,000–22,000; 26,000–
259 30,000) restore both the generative process and the decoder to the baseline configuration, providing
260 within-session reference periods against which each metric’s behavior during a misfit can be
261 calibrated. With this schedule fixed, each diagnostic can be evaluated directly against a known
262 ground-truth fault, and the three responses can be compared across faults that differ in which side of
263 the model is perturbed.

3.2 Simulation Results

Figure 3a shows a realization of the simulation, the decoding results, and each of the three local goodness-of-fit metrics. The top panel shows the true position of the simulated rat in magenta and the predictive decoding distribution in black. The regions with white backgrounds are the periods when the model is correctly specified and each colored background indicates a period of model misspecification. The second panel shows the normalized likelihood of the data as a function of position at the spike times. The third panel shows the spiking from each of the eleven simulated neurons. Since these are sorted by their place field centers, the pattern of spiking aligns with the likelihood as a function of position. Note that the remapped period is the only one where the model misfit leads to incorrect decoding estimates; the other misspecified periods influence the decoding estimates in ways that increase the uncertainty of the estimates but do not lead to inconsistency between the likelihoods and the predictive distributions from the model.

The lower three panels of Figure 3a show, from top to bottom, the values of the HPD overlap, KL divergence, and the rank-based predictive p -value (as $-\log(p)$) for each spike. In each panel a horizontal line marks the threshold used to flag poor fit. For the HPD overlap, where low values indicate misfit, a spike is flagged when its value falls below the threshold, taken as the 1st percentile of the HPD overlap over the initial baseline period. For the KL divergence, where high values indicate misfit, a spike is flagged when its value exceeds the threshold, taken as the 99th percentile over the same baseline period. For the predictive p -value, a spike is flagged when $p \leq 0.05$ (equivalently $-\log(p) \gtrsim 3$), a fixed cutoff rather than a baseline-derived bound. We see that all three metrics agree that there is lack of consistency during the remapping period, where the likelihood from the remapped cells is in a different part of the environment than the model estimate based on the incorrect place field estimates. For the period where history dependence is misspecified, the estimated decoding distribution becomes narrower, but remains consistent with the spike likelihoods, and none of the three metrics flag this period as leading to incorrect estimates. For the period where there is a drift term that is missed by the decoder model, the actual movement trajectory has notably increased velocity, but the decoder is able to keep up, so that the predictive estimates lag behind

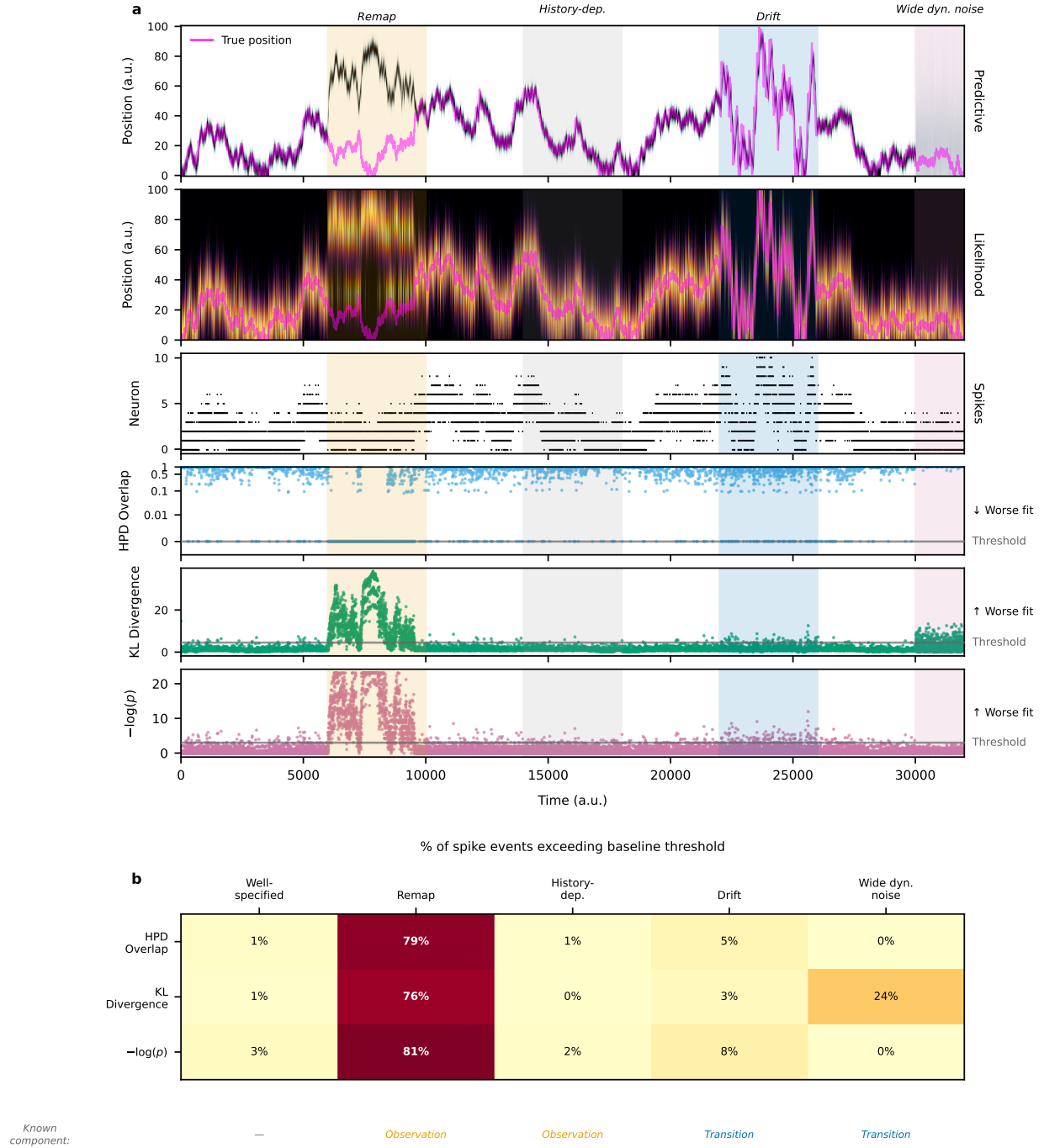


Figure 3: Per-spike goodness-of-fit diagnostics across a four-misfit simulation. The simulation walks through four model misspecifications separated by clean-recovery windows: place-field remapping, unmodeled history-dependent firing, persistent-velocity drift, and wide-dynamics noise. (a) The top block shows the predictive distribution as a heatmap of time and position. The position is overlaid as a magenta line. The misfit time periods are shaded in color. The second from the top block shows the per-spike likelihood as a heatmap, restricted to time bins containing at least one spike. The position is also overlaid as a magenta line. The third block shows the spike raster with neurons sorted by place-field peak position. The bottom three blocks show the three goodness-of-fit diagnostics (HPD overlap, KL divergence, and the rank-based predictive p -value) for each spike event, with a horizontal line marking the threshold used to flag poor model fit. (b) Shows the percent of spike events that are flagged as poor fit by each diagnostic in each phase of the simulation for each metric as well as comparison to the baseline “clean” phases.

but are still consistent with the spike likelihoods. During this period we see a slight increase in the predictive p -value metric, since the predictive distribution tends to lag the actual simulated location, biasing the predictive spiking distribution to neurons to ones that match this lag. For the period where the decoder uses an inflated transition matrix, the predictive density becomes nearly uniform, losing all information from past spiking. Both the HPD overlap and rank-based predictive p -value measures indicate that this broad predictive distribution is consistent with the likelihood of each spike. However, this model misspecification highlights one of the main differences between these two metrics and the KL divergence. When the predictive distribution includes regions that are unlikely to correspond to the observed spike, the KL divergence is substantially increased. This demonstrates that the KL divergence does not quite meet the definition of consistency between distributions that we described above.

Figure 3b shows a heatmap of the fraction of spikes that are flagged by each metric in each period where the model is misspecified. A spike is flagged whenever its diagnostic crosses the corresponding threshold in the direction of worse fit: below the 1st-percentile baseline value for the HPD overlap, above the 99th-percentile baseline value for the KL divergence, or at or below the fixed 0.05 cutoff for the predictive p -value. The heatmap shows a large, consistent fraction of spikes are flagged by all metrics during the remapping period, demonstrating that all three methods are able to identify inconsistent spiking information leading to incorrect state estimates. All three metrics similarly agree in not flagging many spikes during the periods when the model is missing the spiking history dependence and state drift terms. For the period where the decoding model incorrectly specifies the state transition, and therefore overly discounts previous spike information, the HPD overlap and rank-based predictive p -value metrics do not identify inconsistency between the spike likelihoods and the flat predictive distribution, while the KL divergence flags 24% of the spikes. On one hand, this suggests that the KL divergence is sensitive to this true model misspecification; on the other, this suggests that the KL metric will flag spikes whenever the predictive distribution is uninformative, even when that distribution is consistent with the information from the current spike.

4 Real Hippocampal Data Analysis

4.1 Data collection and behavioral task

We examined the ability of these local goodness-of-fit metrics to identify periods of inaccurate decoding from place cells in rat hippocampus as it performed a spatial bandit task on a maze with three radially arranged Y-shaped foraging patches emanating from a center point. A schematic of the maze and the rat's movement in this environment are shown in Figure 4d. The details of this task, and data collection methods are described in detail in Comrie et al. [2024]. Briefly, the rat was trained to run between six reward ports located in three Y-shaped "patches", each with a different probability of reward. The probability of reward periodically switched after a time or trial limit. Twenty-three tetrodes were implanted in the dorsal hippocampus CA1 region. Spikes were detected by a 100 μ V threshold on the recorded neural signal filtered 600 Hz to 6,000 Hz. [Add spike sorting information.]

4.2 Data analysis results

Figure 4 shows a comparison of decoding from this hippocampal spiking data under two state models. The first assumes that the state process moves continuously through space at all times, as would be true if the state only represented the rat's actual location. However, it is known that during periods where the rat is immobile, hippocampal place fields reactivate in sequences that replay past movement trajectories generating a neural representation that can jump to far from the rat's current location [Wilson and McNaughton, 1994, Karlsson and Frank, 2009]. We expect the continuous state model to fail precisely at those times when the neural representation jumps. We compare this with an alternate state model called a continuous-fragmented model that allows the hippocampal representation to jump at discrete times. We have previously used this model structure to capture replay and other nonlocal hippocampal representations [Denovellis et al., 2021].

Figure 4a shows a brief 20 second segment of the decode using the continuous state model. The panels from top to bottom are the same as in Figure 3a. The magenta line shows the rat

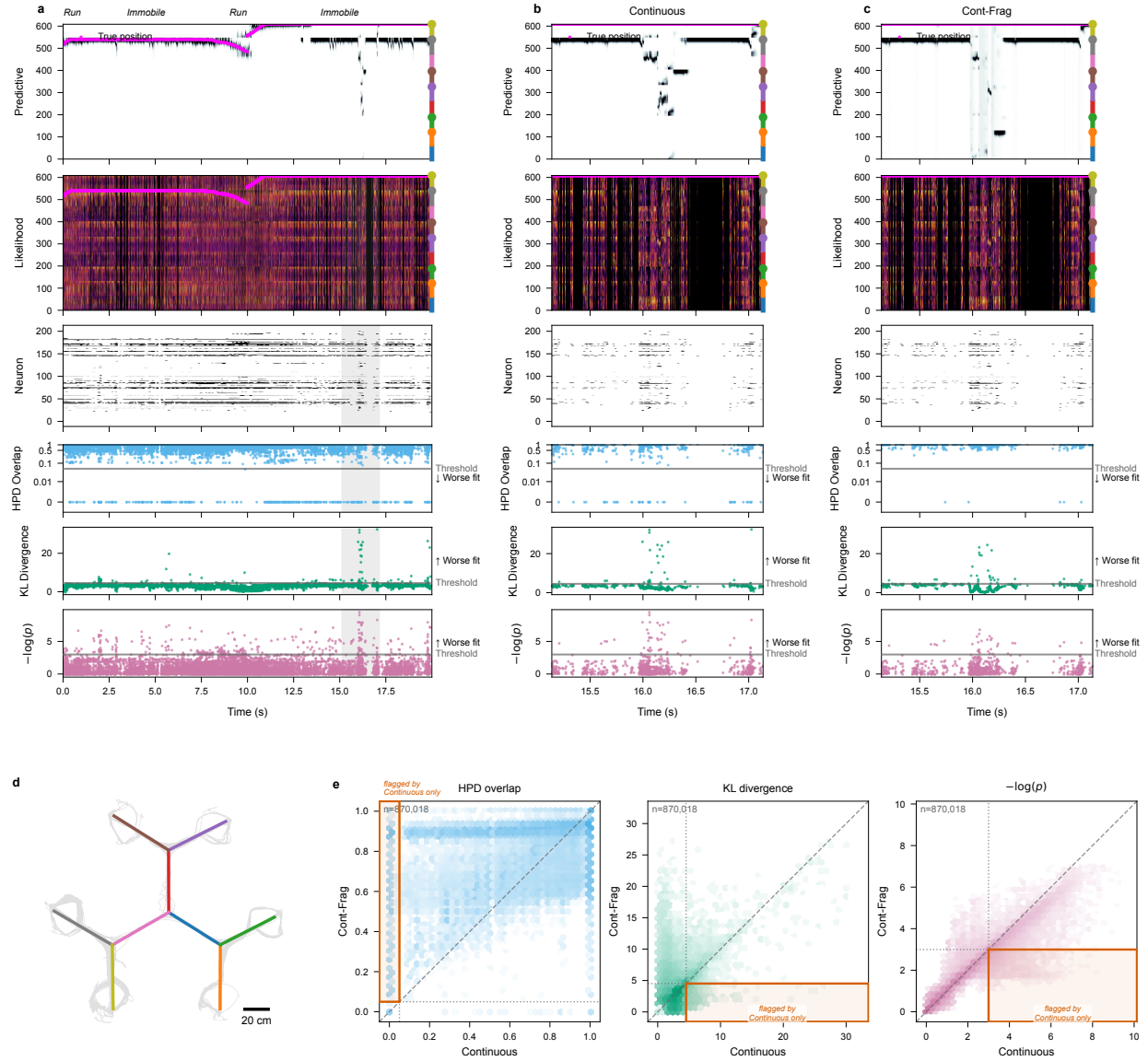


Figure 4: Per-spike goodness-of-fit diagnostics applied to a real hippocampal recording (203 simultaneously recorded cells) during navigation on a multi-arm maze; data from Comrie et al. [2024]. Two state-space decoders—a purely Continuous decoder and a Continuous-Fragmented decoder—are evaluated on the same session. **(a)** Wider context window under the Continuous decoder; behavioral periods (Run / Immobile) are labeled along the top axis, and the shaded band marks the zoom window shown in (b) and (c). **(b), (c)** The same zoom window decoded with the Continuous and Continuous-Fragmented decoders, respectively. Within each column of (a)–(c), all rows share a common time axis (x , seconds, relative to the start of the context window). **Row 1, Predictive:** heatmap of the predictive posterior with linearized position on y and the animal's true linearized position overlaid. **Row 2, Likelihood:** same axes, restricted to time bins containing at least one spike, showing the combined per-spike likelihood. **Row 3, Neuron:** spike raster with neurons on y , sorted by place-field peak position. **Rows 4–6 (HPD overlap, KL divergence, $-\log(p)$):** one marker per spike event placed at its spike time, with the per-spike diagnostic value on y ; a horizontal line marks the threshold used to flag poor model fit (HPD overlap and the predictive p -value at a fixed 0.05 cutoff; KL divergence at the 99th-percentile baseline threshold carried over from the simulation in Figure 3, ≈ 4.52), and the direction of worse fit is labeled at the right. **(d)** 2D layout of the track with the animal's trajectory overlaid in grey and a 20 cm scale bar. **(e)** Pairwise comparison of per-spike diagnostic values between the two decoders across the whole session, one hexbin panel per metric (HPD overlap, KL divergence, $-\log(p)$); x -axis is the Continuous decoder, y -axis is the Continuous-Fragmented decoder, and the dashed line is identity. Dotted lines mark the per-metric flag threshold on each axis (the same values used in panels (a)–(c)), and the outlined box marks the quadrant of spikes flagged by the Continuous decoder but not the Continuous-Fragmented decoder. Hexagons are colored by spike-event density (log scale).

running into one of the maze arms (the discontinuity in the magenta line represents a transition into an arm segment) and then becoming immobile while it rests. At around 12.5 seconds into the segment, while the rat is immobile, the decoded representation shifts to a new location, and then at around 16 seconds into the segment (highlighted with a gray background in lower panels), we find a classic replay event with rapid transitions between areas of the maze. During this replay event, the continuous state model is unable to keep up and the predictive distribution consistently lags behind the information coming from the likelihood of the observed spikes. In the lower three panels, we see many of the spikes flagged as inconsistent with the predictive model estimates by all three goodness-of-fit metrics. Figure 4b shows a zoom-in of the highlighted window containing the replay event.

The contrast between decoding with the continuous and continuous-fragmented state models is shown in Figures 4b and c. We see that the likelihood for each spike is identical for these models (second panel from top) but the continuous-fragmented model decodes a rapidly moving state trajectory that the continuous model is unable to track. This happens because the predictive distribution becomes uniform or nearly uniform at time steps where the representation is estimated to be fragmented and the likelihood of subsequent spikes can then cause a jump in the state estimate. For the HPD overlap metric (fourth panel from top, blue), we see that nearly all of the spikes flagged in the continuous state model are no longer flagged when using the continuous-fragmented model that can track rapid jumps in the state process. For the rank-based predictive p -value (bottom panel, pink), we see that some of the $-\log(p)$ values are reduced for the continuous-fragmented relative to the continuous state model. This is because the continuous state model leads to predictive densities that are concentrated in the wrong location, which makes the observed spiking very unlikely, while the continuous-fragmented state model leads to broad predictive distributions during replay events, which makes the observed spikes still unlikely but less so. For the KL divergence metric (fifth panel from top, green), we find that many spikes retain high values for the continuous-fragmented model. This is consistent with our simulation results, where nearly flat predictive distributions led to increased KL values, even though they are consistent with the spiking likelihood according to our

earlier definition.

Figure 4e shows scatterplots comparing the values of each of the local goodness-of-fit metrics for the continuous state model versus the continuous-fragmented state model applied to each spike. The leftmost panel shows the values of the HPD overlap metric in blue. Here, points above the 45° line indicate spikes for which the continuous-fragmented state model produces predictive distributions more consistent with the spiking data than the continuous state model. Of note here is the linear cloud of points running vertically at $x = 0$ and $y > 0$, which indicate spikes for which the predictive distribution from the continuous state model has no overlap with the spike likelihood but for which the predictive distribution from the continuous-fragmented state model has nonzero overlap. There is no corresponding cloud at $y = 0$ and $x > 0$, suggesting that the HPD overlap is successful at identifying times when the continuous state model is inconsistent with the observed spiking and that adding the fragmented state process into the model alleviates this inconsistency.

The center panel of Figure 4e shows values of the KL divergence in green. In this case, points below the 45° line indicate spikes for which the continuous-fragmented state model produces predictive distributions more consistent with the spiking data than the continuous state model. There is indeed a large concentration of points below this line in the region where the value for the continuous model is below 5. More striking though is the region above the 45° line where the value for the continuous model is around 2 and the value for the continuous-fragmented model goes up to 20 or so. As seen in the simulation, this corresponds to situations where the predictive distribution from the continuous-fragmented model is broad, as occurs whenever a fragmented state is estimated. This suggests that the KL divergence is less useful in identifying inconsistency from the continuous-fragmented model.

The right panel of Figure 4e shows the value of the negative log of the rank-based predictive p -value ($-\log(p)$) in pink. Once again, points below the 45° line indicate spikes for which the continuous-fragmented state model produces predictive distributions more consistent with the spiking data than the continuous state model. Of note here is the cloud of points below this line that extends beyond a value of 3 on the x -axis (corresponding to a p -value below 0.05, since $-\log(0.05) \approx 3$).

There are no corresponding spikes above the line with a value for the continuous-fragmented model above 3. As with the HPD overlap metric, this suggests that the rank-based predictive p -value is also successful at identifying times when the continuous state model is inconsistent with the observed spiking.

5 Discussion

State space modeling has become a powerful tool for describing the latent dynamics of neural systems. However, validating these models is inherently difficult because latent states are unobservable and many models are purposefully simplified. To address this, we developed multiple local goodness-of-fit metrics that compare the instantaneous information received from each spike to the aggregated information computed based on recent spiking activity and the state-space model assumptions. Rather than dismissing a model in its entirety, these metrics are able to assess the consistency between immediate observations and the information accumulated through the state-space structure, providing a more granular assessment of decoding reliability.

We defined three classes of metrics: F-divergence measures (e.g., KL divergence), overlap ratios between credible regions (HPD overlap), and predictive checks. Although KL divergence is computationally simple, it is sensitive to distribution tails and lacks a clear upper bound to assess similarity. In contrast, the HPD overlap offers a more robust geometric measure of similarity, consistently maintaining lower error rates across various conditions. Predictive checks provide a direct statistical assessment through observation-level p -values, but are more computationally intensive. Although each metric has particular advantages, when used in combination, they provide a more complete perspective on decoding reliability.

We illustrated the application of these metrics to simulated and real spike-sorted data from rat hippocampus. These same metrics naturally extend to other neural data types when incorporated into state space models. In particular, recent work has shown that clusterless neural spiking models can substantially reduce bias and variance in neural decoding and estimation problems when compared

to spike-sorted models [Deng et al., 2016]. The metrics, as defined in Section 2, naturally extend to clusterless models. In that case, a lack of fit for any particular spike suggests that a spike with that waveform was unexpected based on the accumulated information from the state space model up to that point in time.

For simplicity, in this paper, we focused on comparing instantaneous information in the likelihood of each spike with aggregated information from the past contained in the one-step prediction density. A natural and powerful extension of these methods is obtained by replacing the one-step prediction density with any posterior density derived from the state-space framework. In particular, replacing the one-step prediction density with a smoothing density (e.g., from a fixed-interval smoother over the observation period) would allow us to leverage both past and future data to increase statistical power in identifying model misspecification.

One potential challenge for applying these methods broadly has to do with the computational burden of computing these metrics as the dimensionality of the state process increases. In all of our examples, we computed the KL divergence measure and the HPD overlap by partitioning the state space into a lattice and computing the likelihoods and prediction densities at each point in that lattice. As the dimension of the state increases, the size of that lattice will grow exponentially. While the predictive checks are the most computationally burdensome in lower state dimensions, the fact that we are sampling from the state-space rather than computing values over a lattice means that in higher dimensions the predictive checks will become more computationally efficient. It also suggests that we may be able to improve the efficiency of the KL and HPD overlap metrics by using sampling-based estimation methods. This will not solve the curse of dimensionality, as the number of samples required for accurate estimates will likely still grow with dimension, but it is likely to allow for efficient estimation in moderate dimensions. Another approach for dealing with this issue may be to compute these consistency metrics in lower-dimensional subspaces of the state space. Identifying subspaces that retain the statistical power of the original space is a topic for future exploration.

Another potential challenge relates to identifying time-scales for which to assess for lack of fit.

These methods allow us to focus on time-scales as short as individual spikes, and for our place cell decoding examples individual spikes are sufficiently informative that we are often able to detect local model misfit at this time-scale. For other neural systems, it may be the case that local model errors are more likely to be detected over longer time-scales. In this paper, we discuss how to extend these goodness-of-fit measures to longer time intervals, but determining what time-scales are most appropriate may require relying on prior knowledge about the system, or exploring the values of these metrics across multiple time-scales.

We believe that these local goodness-of-fit tools provide an important step toward making inferences from state-space neural models more reliable and interpretable. By pinpointing specific moments when the model assumptions break down, researchers can be more confident in their inferences. Ultimately, these measures empower us to refine our understanding of neural dynamics and deepen our understanding of how the brain encodes and processes information.

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References

- A. Bhattacharyya, S. Gayen, K. S. Meel, D. Myrisiotis, A. Pavan, and N. V. Vinodchandran. Total variation distance meets probabilistic inference, 2023. arXiv preprint arXiv:2309.09134.
- A. E. Comrie, E. J. Monroe, A. E. Kahn, E. L. Denovellis, A. Joshi, J. A. Guidera, T. A. Krausz, J. D. Berke, N. D. Daw, and L. M. Frank. Hippocampal representations of alternative possibilities are flexibly generated to meet cognitive demands. bioRxiv, 2024. URL <https://www.biorxiv.org/content/10.1101/2024.09.23.613567v1>.
- T. F. Crack. A note on Karl Pearson’s 1900 chi-squared test: two derivations of the asymptotic

distribution, and uses in goodness of fit and contingency tests of independence, and a comparison with the exact sample variance chi-square result, 2018. Available at SSRN 3284255.

X. Deng, D. F. Liu, K. Kay, L. M. Frank, and U. T. Eden. Clusterless decoding of position from multiunit activity using a marked point process filter. *Neural computation*, 27:1438–1460, 2015.

X. Deng, D. F. Liu, M. P. Karlsson, L. M. Frank, and U. T. Eden. Rapid classification of hippocampal replay content for real-time applications. *Journal of neurophysiology*, 116:2221–2235, 2016.

E. L. Denovellis, A. K. Gillespie, M. E. Coulter, M. Sosa, J. E. Chung, U. T. Eden, and L. M. Frank. Hippocampal replay of experience at real-world speeds. *Elife*, 10:e64505, 2021.

María Jesús Bayarri García and ME Castellanos. Bayesian Checking of the Second Levels of Hierarchical Models. *Institute of Mathematical Statistics*, pages 363–367, 2007.

A. Gelman, X. L. Meng, and H. Stern. Posterior predictive assessment of model fitness via realized discrepancies. *Statistica sinica*, pages 733–760, 1996.

Irwin Guttman. The use of the concept of a future observation in goodness-of-fit problems. *Journal of the Royal Statistical Society: Series B (Methodological)*, 29:83–100, 1967.

T. Hastie, R. Tibshirani, and J. Friedman. The elements of statistical learning, 2009.

W. C. Hsin, U. T. Eden, and E. P. Stephen. Switching Functional Network Models of Oscillatory Brain Dynamics. In *2022 56th Asilomar Conference on Signals, Systems, and Computers*, pages 607–612, 2022.

L. M. Jones, A. Fontanini, B. F. Sadacca, P. Miller, and D. B. Katz. Natural stimuli evoke dynamic sequences of states in sensory cortical ensembles. *Proceedings of the National Academy of Sciences*, 104:18772–18777, 2007.

Mattias P Karlsson and Loren M Frank. Awake replay of remote experiences in the hippocampus. *Nature neuroscience*, 12(7):913–918, 2009.

493 Solomon Kullback and Richard A Leibler. On information and sufficiency. *The annals of*
494 *mathematical statistics*, 22:79–86, 1951.

495 X. L. Meng. Posterior predictive p -values. *The annals of statistics*, 22:1142–1160, 1994.

496 Gemma E Moran, David M Blei, and Rajesh Ranganath. Population predictive checks, 2019. arXiv.

497 K. B. Newman, S. T. Buckland, B. J. T. Morgan, R. King, D. L. Borchers, D. J. Cole, P. Besbeas,
498 O. Gimenez, and L. Thomas. Modelling population dynamics. *Methods in Statistical Ecology*,
499 2014.

500 X. Shen and J. Zhuang. Residual analysis-based model improvement for state space models with
501 nonlinear responses. *IEEE Transactions on Emerging Topics in Computational Intelligence*, 8:
502 1728–1743, 2024.

503 M. Stone. Cross-validators choice and assessment of statistical predictions. *Journal of the royal*
504 *statistical society: Series B (Methodological)*, 36:111–133, 1974.

505 Long Tao, Karoline E Weber, Kensuke Arai, and Uri T Eden. A common goodness-of-fit framework
506 for neural population models using marked point process time-rescaling. *Journal of computational*
507 *neuroscience*, pages 147–162, 2018.

508 A. Vinograd, A. Nair, J. H. Kim, S. W. Linderman, and D. J. Anderson. Causal evidence of a line
509 attractor encoding an affective state. *Nature*, 634:910–918, 2024.

510 Matthew A Wilson and Bruce L McNaughton. Reactivation of hippocampal ensemble memories
511 during sleep. *Science*, 265(5172):676–679, 1994.

512 S. Zeng and U. T. Eden. A state-space framework for causal detection of hippocampal ripple-replay
513 events. *IEEE Transactions on Biomedical Engineering*, 2025.